Introduction to Sets

MTH 317H — Week 1

Name: _____

1 What is a set?

Definition 1.1. A *set* is a well-defined collection of objects. The objects in a set are called its *elements*.

Preparation

Three features of sets are important.

1. **Unordered:** changing the order does not change the set:
$$\{1, 2, 3\} = \{2, 3, 1\}.$$
2. **No repetition:** listing an element more than once does not create a new element:
$$\{\heartsuit, 3, \text{blue}\} = \{\heartsuit, \heartsuit, 3, \text{blue}\}.$$
3. **Well-defined:** it must be possible to decide whether an object belongs to the collection.

For example, "the colors of the rainbow" describes an objective collection. By contrast, "our favorite colors" does not specify a set until the people involved and their choices are fixed.

2 Set notation

We list the elements of a set inside curly braces. Sets are usually named with capital letters. For example,
$$X = \{x_1, x_2, x_3\}.$$
The symbol $\in$ means "is an element of," while $\notin$ means "is not an element of." Thus
$$x_1 \in X, \qquad x_4 \notin X.$$

Definition 2.1. For sets A and B, we write $A \subseteq B$ when every element of A is also an element of B. In that case, A is a *subset* of B.

Preparation

Keep elements and subsets separate:

$x \in A$ means x is an element of A, while $\{x\} \subseteq A$ means $\{x\}$ is a subset of A.

3 Standard sets of numbers

The following sets occur throughout mathematics:

Name	Notation and description
Empty set	$\emptyset = \{\}$, the set with no elements.
Natural numbers	$\mathbb{N} = \{1, 2, 3, 4, \ldots\}$.
Integers	$\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$.
Rational numbers	$\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z},\ q \neq 0 \right\}$.
Real numbers	$\mathbb{R}$, the points on the real number line.
Complex numbers	$\mathbb{C} = \{a + bi : a, b \in \mathbb{R}\}$.

These sets satisfy the chain of inclusions

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}.$$

4 Describing a set

There are two common ways to describe a set.

4.1 Roster notation

In *roster notation*, we list the elements:

$$X = \{x, y, z\}, \qquad E = \{\ldots, -4, -2, 0, 2, 4, \ldots\}.$$

4.2 Set-builder notation

In *set-builder notation*, we state the rule that determines membership:

$$E = \{2n : n \in \mathbb{Z}\} = \{x \in \mathbb{Z} : x \text{ is even}\}.$$

The colon is read "such that." A vertical bar may be used instead:

$$\{2n : n \in \mathbb{Z}\} = \{2n \mid n \in \mathbb{Z}\}.$$

Proof

The roster and set-builder descriptions of E give the same set. Every number displayed in the roster is twice an integer, so it belongs to $\{2n : n \in \mathbb{Z}\}$. Conversely, if $x = 2n$ for an integer n, then x is an even integer and therefore appears in the roster. Thus the two descriptions have exactly the same elements.

5 Cardinality

Definition 5.1. The *cardinality* of a set X, written $|X|$, is the number of elements in X.

For the finite set $X = \{x, y, z\}$, we have $|X| = 3$. The empty set has cardinality $|\emptyset| = 0$. The set of even integers is infinite; following the informal notation in the source notes, one may write $|E| = \infty$.

Remark 5.2. The braces matter: $\emptyset$ has no elements, whereas $\{\emptyset\}$ has one element. Hence $|\emptyset| = 0$ but $|\{\emptyset\}| = 1$.

6 Set operations

Let A and B be sets contained in a universe U.

Preparation

The basic set operations are characterized by membership:

$$\begin{aligned} A \cup B &= \{x : x \in A \text{ or } x \in B\}, \\ A \cap B &= \{x : x \in A \text{ and } x \in B\}, \\ A \setminus B &= \{x : x \in A \text{ and } x \notin B\}, \\ A^c &= U \setminus A, \\ A \times B &= \{(a, b) : a \in A \text{ and } b \in B\}. \end{aligned}$$

The union combines the elements of the two sets, and the intersection keeps only the elements shared by both sets. The difference $A \setminus B$ removes the elements of B from A. A complement is always taken relative to a specified universe. The Cartesian product is a set of ordered pairs, so in general $(a, b) \neq (b, a)$.

Example 6.1 (Computing the operations)**.** Take

$$A = \{2, 4, 6\}, \qquad B = \{4, 8, 12\}, \qquad U = A \cup B.$$

Then

$$\begin{aligned} A \cup B &= \{2, 4, 6, 8, 12\}, \\ A \cap B &= \{4\}, \\ A \setminus B &= \{2, 6\}, \\ B^c = U \setminus B &= \{2, 6\}. \end{aligned}$$

Also,

$$\begin{aligned} A \times B = \{&(2, 4), (2, 8), (2, 12), (4, 4), (4, 8), (4, 12), \\ &(6, 4), (6, 8), (6, 12)\}. \end{aligned}$$

Since $|A| = 3$ and $|B| = 3$, the product has $|A \times B| = |A|\,|B| = 9$ elements.

Summary

Notation	**Meaning**
$x \in A$	x is an element of A.
$A \subseteq B$	Every element of A is also in B.
$\lvert A \rvert$	The number of elements in A.
$A \cup B$	Elements in A or B (or both).
$A \cap B$	Elements in both A and B.
$A \setminus B$	Elements in A but not in B.
A^c	Elements of the universe that are not in A.
$A \times B$	Ordered pairs (a, b) with $a \in A$ and $b \in B$.